PROJECT-BASED LEARNING

PHASE-II EVALUATION

Ambulance and Emergency dispatch system

Team ID: DSCPP-III-2026-T148

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T148
Semester: 3rd
Section: K,E,ML1,ML3
Domain: *Btech CSE*

Team Members

1. *Neel Hajariwala -2510011575*
2. *Keshav Namdev -2510360097*
3. *Lichhavi Chauhan -2510370231*
4. *Aditya Ahlawat -2512510001*

Mentor

Mr.Ramesh Singh Rawat

Interactions completed: _____

Brief Discussion:C++ delivers high-performance execution for instant pathfinding, file-based logging ensures lightweight zero-overhead storage, and Agile methodology enables rapid, iterative testing of critical emergency logic.

Phase-I Recap & Feedback Incorporated

Phase-I Status

During Phase-I, we proposed the basic architecture of the Ambulance and Emergency Dispatch System and planned the emergency triage process. We completed the initial project scoping, system design, prototype planning, module structuring, and dataset organization, establishing a foundation for the next phase of development.

Feedback Received

- *Select the most suitable ambulance and reduce response time.*
- *Test the Max-Heap using multiple emergency requests*

Changes Incorporated

- *Add a hospital management module to check bed availability*
- *notifications to inform drivers about emergency location.*

Problem Statement & Finalized Objectives

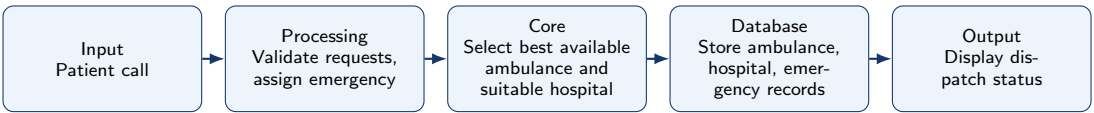
Problem Statement

The finalized problem addressed by the project is the delay and inefficiency in manual ambulance dispatch during emergencies. The system will use C++ and Data Structures to prioritize emergency requests based on urgency, identify and assign the nearest suitable available ambulance, and manage ambulance status efficiently. It aims to reduce response time, improve resource utilization, and ensure timely ambulance allocation through a structured and reliable dispatch process.

Finalized Objectives

1. *To identify and assign the nearest available ambulance based on emergency priority and distance.*
2. *To prioritize emergency requests and handle ambulance acceptance or rejection efficiently.*

Updated System Architecture / Workflow



Major Components

- *Ambulance Availability and Management*
- *Emergency Priority and Dispatch Algorithm*
- *Hospital Bed and Patient Management*

Changes Since Phase-I

- *Integrated modular C++ system architecture.*
- *Hospital bed availability and alternate hospital selection.*
- *Automated ambulance selection using priority and distance, with driver rejection handling.*

Module Design & Functional Requirements

Major Modules

1. *Emergency Request Management Module*
2. *Ambulance Management Module*
3. *Dispatch and Route Optimization Module*
4. *Hospital Management Module*

Functional Requirements

- *The system shall accept emergency requests and assign priority based on severity.*
- *The system shall allow drivers to log in and update ambulance availability status.*

Module Dependency

Emergency details and priority, along with ambulance availability, are used for ambulance selection and route planning. Hospital bed availability helps determine the suitable hospital.

Technology Stack & Development Methodology

Technology Stack

- Programming: *C++*
- Database: *MySQL*
- Framework: *STL*
- Development Tool: *VS code*
- Version Control: *GitHub*

Development Methodology

1. Emergency Triage Scoping
2. Heap Logic
3. C++ core coding
4. Dijkstra Path Validation
5. Performance Optimization

Brief Discussion:C++ delivers high-performance execution for instant pathfinding, file-based logging ensures lightweight zero-overhead storage, and Agile methodology enables rapid, iterative testing of critical emergency logic.

Implementation of PBL Course Concepts

Course	Concepts Actually Implemented	Project Module
Data Structures in C	*Vectors, priority queues,unordered maps,Dijkstra's*	*Triage Engine*
OOPs with C++	*Classes, objects, structures, and encapsulation.*	*Dispatch Manager*

Brief Discussion:C++ delivers high-performance execution for instant pathfinding, file-based logging ensures lightweight zero-overhead storage, and Agile methodology enables rapid, iterative testing of critical emergency logic.

Development & Implementation Progress

Module	Status	Work Completed
Ambulance Management	Completed	*Driver status and availability tracking*
Emergency Dispatch	Completed	*Priority queue and ambulance selection*
Hospital Management	In Progress	*alternate hospital selection*
Route Optimization	Pending	*Dijkstra's shortest-path implementation*

Overall Project Completion: *65%*

Working Prototype / Demonstration

Implemented Features

- *Emergency Request Management: Register and prioritize emergency calls.*
- *Ambulance Availability Tracking: Identify available and busy ambulances.*
- *Smart Ambulance Dispatch: Select the nearest suitable ambulance and handle driver rejection.*
- *Hospital Coordination: Check bed availability and select an alternative hospital when needed.*

Prototype / Output

```
AMBULANCE & EMERGENCY DISPATCH
Patient: Test Patient
Emergency: Medical Emergency

Step 1: Emergency Prioritization
  Priority Level: 4 (Critical, first)
Step 2: Route Calculation
  Road network built using Dijkstra
Step 3: Nearest Ambulances
  Ambulance 1 - 4 km
  Ambulance 2 - 5 km
  Ambulance 3 - 8 km
  Ambulance 4 - 10 km
Step 4: Ambulance Assignment
  Assigned Ambulance ID: 2 (5 km)
  Status: Assigned
  Double assignment blocked
Step 5: Hospital Selection
  City Hospital - 3 km
  Available Beds: 5
```

Testing, Results & Validation

Test / Module	Expected Result	Observed Result	Status
Priority Queue	*Critical first*	*Critical prioritized*	Pass
Route Calculation	*Shortest route*	*Route calculated using Dijkstra*	Pass
Ambulance Dispatch	*Assign eligible ambulance*	*Ambulance assigned; duplicate blocked*	Pass

Results / Validation

The C++ DSA prototype successfully demonstrated emergency prioritization, shortest-route calculation, ambulance assignment, and hospital selection. The web dashboard also displayed sample emergency and hospital results. Testing used simulated data; live GPS and real-time ambulance tracking remain future work.

GitHub Repository & Development Activity

Repository Information

- GitHub Link: *https://github.com/LICHHAVI26/DSCPP-PBL*
- README.md: *https://github.com/LICHHAVI26/DSCPP-PBL/tree/main*
- Contributors: *All Members Added*
- Repository Status: *Updated*

Development Activity

- *C++ DSA modules developed and integrated*
- *GitHub commits maintained with meaningful messages*
- *Individual team contributions documented*
- *Latest implementation uploaded to GitHub*

Individual Contribution

Member	Role	Technical Contribution	GitHub Commits
Neel Hajariwala	*Lead*	*Project planning, system architecture, coordination and module integration*	*06*
Keshav Namdev	*Developer*	*Max-Heap/Priority Queue, Dijkstra algorithm and C++ implementiond*	*06*
Lichhavi Chauhan	*Developer/Docs*	*Workflow, flowchart, module development and documentation*	*07*
Aditya Ahlawat	*DB/Testing*	*Data management, ambulance status/logs, testing and validation*	*03*

Individual Evaluation

Every member should be prepared to explain and demonstrate their own technical contribution to the project.

Challenges, Remaining Work & Phase-III Roadmap

Challenges Faced

- Technical: Implementing Dijkstra's algorithm and emergency priority queues.
- Integration: Connecting C++ modules with the web dashboard.
- Implementation: Managing ambulance assignment and preventing duplicate dispatch.

Solutions / Learning

- Fixed compilation and integration errors through debugging.
- Applied DSA for priority-based dispatch and shortest-route calculation.

Phase-III Roadmap

1. Complete driver details and ambulance status modules.
2. Integrate C++ modules with the web dashboard.
3. Test dispatch logic and improve performance.
4. Validate the system using multiple test cases.
5. Complete documentation and final demonstration.

Expected Outcomes & References

Expected Final Outcomes

- *Complete web-based ambulance dispatch prototype using C++ and DSA*
- *Priority-based emergency handling and Dijkstra shortest-route calculation*
- *Nearest-four ambulance selection and duplicate-assignment prevention*
- *Hospital selection based on distance and bed availability*
- *Potential for future GPS integration and real-world deployment*

References

1. *Dijkstra, E. W. (1959). A Note on Two Problems in Connexion with Graphs. Numerische Mathematik, 1, 269–271.*
2. *Cormen et al. Introduction to Algorithms. MIT Press.*
3. *Lafore, R. Data Structures and Algorithms in C++.*
4. *C++ Reference — Priority Queue: https://en.cppreference.com/w/cpp/container/priori*

Thank You
Questions & Suggestions